

Project Proposal: Accelerating hybrid quantum-classical machine learning tasks on HPC platforms.

Maha A. Metawei

Introduction

A topic-aware classifier based on a hybrid quantum-classical model is developed as part of my PhD thesis [1] to classify sentences based on their corresponding topic meaning. The model was trained using 100 pairs of sentences belonging to different topics: programming and cooking. Quantum computing is helping to reduce the computational cost of many problems. For instance, a quantum computational advantage could be achieved over certain classical Natural Language Processing (NLP) models using currently available quantum computing platforms.

In our previously mentioned work, an enhancement in classification accuracy was achieved on noiseless and noisy classical simulation over classical tensor-network model. The article in [1] goes through details of different Quantum Natural Language Processing (QNLP) experiment stages while explaining the uniqueness of a QNLP experiment. In terms of problem size, the QNLP stands out among other types of classification experiments. When the word meaning dimension is increased, the problem size grows exponentially, which affects the classification process. Another factor of scalability is the sentence length. For simple affirmative sentence, there are two pregroup types: noun and sentence. Depending on how many qubits are assigned to each pregroup type, the number of qubits on which a sentence's corresponding quantum circuit is constructed, or the circuit width, will scale as the sentence's length does.

Problem Statement

As expected from variational quantum circuits [2], the tested QNLP model showed resilience to noise when tested on a classical simulator with induced noise model. On the other hand, **the QNLP model training typically takes more than one hour to complete on a CPU based simulation to process an around six words sentence.** The proposed prototype model in my thesis was only tested as a proof of concept on a CPU-based quantum simulator. Because the model experienced performance challenges during training, **employing the HPC machines on the host institute's premises can provide a workaround to this challenge.**

Background

Despite the revolutionary potential of quantum computers, current hardware poses significant limitations. Qubit counts remain modest, susceptible to noise and prone to errors, hindering the complexity of problems they can tackle. This is a challenge known as Algorithms-to-Machine gap [3]. Additionally, controlling and manipulating qubits presents a challenge, leading to limited functionality and increased error rates. These hardware limitations restrict the practical applications of quantum computers to niche areas, requiring substantial advancements before achieving the breakthroughs promised by its theoretical capabilities. While early results are promising, overcoming these hardware hurdles is crucial for unlocking the true power of quantum computing and ushering in its transforming era.

As a workaround, hybrid quantum-classical algorithms are significant in solving real-world problems today as they can provide a boost to the quantum device capability [4, 5, 6, 7].

Objectives

- Testing the QNLP model's capability under various conditions, including experimental environment, input vocabulary, and sentence length.
- Enhancing hybrid quantum-classical machine learning tasks performance and results with two different approaches: CPU and GPU-based simulation.
- Learning the best practices for installing and upgrading GPU-based quantum simulators on HPC platforms and transferring this expertise to my home institution.
- The project is consistent with my current institution's long-term research plans in High Performance Computing.
- Connecting with High Performance Computing experts in the host institutions to facilitate future collaboration and support.

Significance

Acquiring quantum computing technology is a significant barrier for economies with limited resources in Africa and Asia. However, this technology offers an excellent potential to provide efficient solutions to problems in essential sectors such as finance, healthcare, and agriculture. Running quantum simulations on High Performance Computing (HPC) platforms is an ideal alternative that provides a practical and cost-effective method for developing such quantum solutions.

Methodology

HPC-backed Quantum Simulation:

The implemented prototype was built on the TKet quantum simulator platform from Qantinnuum, which was best suited for the lambeq [8] package used to parse sentences and convert them to quantum circuits. Because the lambeq package can be interfaced with the Qiskit simulator, the first step in improving performance is to implement and test the model with the Qiskit Aer simulator (without GPU support).

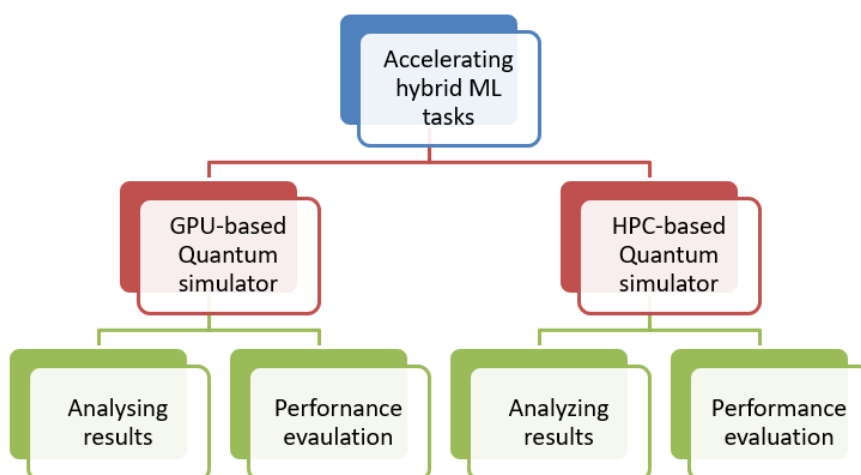


Figure 1: Accelerating hybrid jobs

GPU-based Quantum Simulation:

Classical simulation of quantum circuits requires heavy computational resources. One direction to classically simulate larger circuits is using a GPU-based quantum simulator like for example cuQuantum [9]. These simulators are especially suited for variational algorithms used by quantum machine learning models, which can benefit from improved methods such as the ad-joint differentiation method. As cuQuantum is Python-based, it integrates well with many machine learning packages. Using the Qiskit [10] Aer simulator's GPU-based version and repeating the experiments on the GPU-based platform is an additional strategy for enhancing performance.

Analysis & Testing

Analyzing the extracted results is a crucial part of the project. The analysis shall test the effectiveness of the used acceleration approach based on the following metrics:

- Classification accuracy values
- Runtime of the training process

Planned experiments to be conducted:

The proposed experiments will test the same model using different computing platforms. Based on the available resources and time, other variations to the training dataset might be tested like the size of vocabulary, the sentence length, or translating to other languages.

Table 1: Planned experiments to be run.

Experiment ID	Platform	#Qubits	Running device	Time	Accuracy
Experiment 1	Qiskit	TBD	CPU	-	-
Experiment 2	Qiskit	TBD	GPU	-	-
Experiment 3	cuQuantum	TBD	GPU	-	-

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